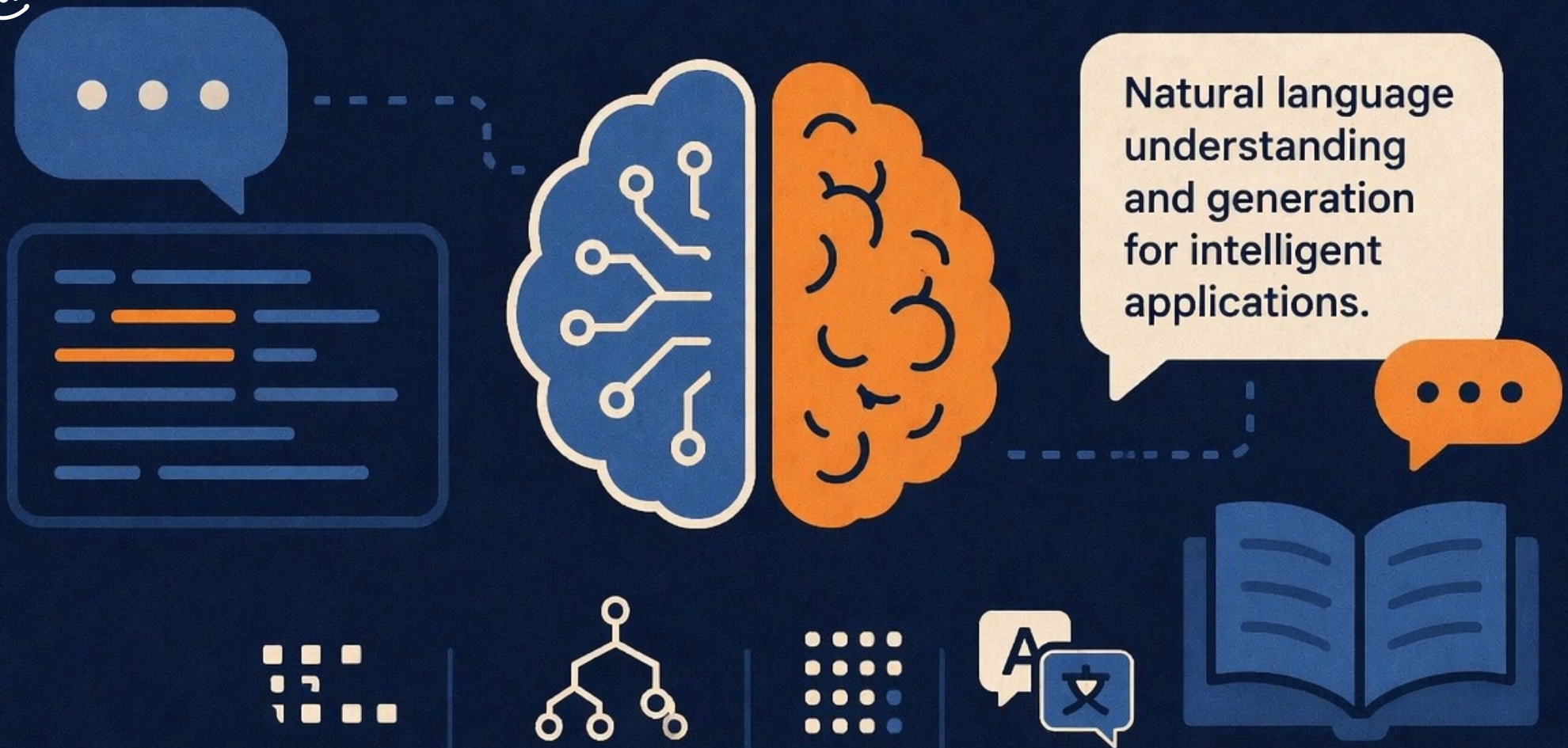
NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



Lecture: Text Pre-processing

#Agenda

1) Why Text processing?
2) Different types of pre-processing
3) Python libraries (NLTK & Spacy)
4) UNET

"What is Machine learning? Who created this domain? Branches of ML?"

1) Text auto-completion
2) Text classification
3) Sentiment Analysis
4) Topic Modelling
5) NER

Text-preprocessing

1) Cleaning ✓
2) Normalization ✓
3) Tokenization ✓

Input
1) Audio (speech)
2) Type
3) Image data / video

Saved in diff format.
(file)

.txt, .pdf, .docx, .html, .md, .parquet
.csv, .xlsx, .yml.

NLTK

Spacy

This is a ML class I want to go home after this

UNET

Image (Input) ⟶ Segmented Mask (Output)

Architecture



Encoder (Downsampling) ⟶ Bottleneck ⟶ Decoder (Upsampling)

1) Encoder

1) Extract features (Find out whats in the image)

2) Reduces spatial size

↳ Conv, ReLu, Maxpooling

2) Decoder

a) Restroes spatial resolution (Find out where it is located)

↳ Transposed Conv

↳ Feature reconstruction

3) Skip Connections

a) Connect the encoder layers to decoder layers

b) passing the fine grain detail directly

w/o this → Blurry segmentation

w this → sharp segmentation